The Likert Blindspot: Measurement-Mode Divergence in LLM Judge Style Evaluation

Anonymous ACL submission

Abstract

Likert and pairwise scoring yield divergent style-sensitivity profiles in LLM judges. We define the *Likert Blindspot*: statistically significant pairwise sensitivity absent from Likert, admitting non-exclusive explanations (measurement attenuation, format amplification, construct divergence). We introduce Controlled Style Decomposition (CSD), a black-box diagnostic combining NLI-gated content control, per-axis style decomposition, and position-controlled pairwise comparison. Across four target judges and two axes (formality, verbosity), pairwise reaches significance ($p < 0.001$) where Likert does not ($p > 0.28$), selectively replicated across three benchmarks and confirmed by seven statistical methods; a fifth judge shows benchmark-dependent sensitivity. At $n = 355$, Likert effect sizes collapse toward zero while pairwise remains stable; at $n = 330$, Likert begins detecting only the strongest effects, revealing a 3–4× efficiency gap rather than absolute failure. Generalized estimating equation (GEE) decomposition shows per-axis confound structures: verbosity is style-dominated (Style OR 8–13 at n=86–88), while formality reflects joint style–quality confounding. Non-tie concordance (76.6%, $p < .001$) favors attenuation over construct divergence. Likert-only audits risk missing style preferences that propagate through pairwise-based alignment pipelines[1].

1 Introduction

"What is the average height for a male?" In our experiments, given a formal and a casual answer with verified identical content (NLI entailment = 0.993), GPT-4o rates the casual variant *higher* on a 1–10 Likert scale (8 vs. 5), yet consistently selects the formal variant in pairwise comparison. Same judge, same texts, opposite conclusions.

[1]Code and data are available at `https://github.com/anonymous-nlp-2026-2/causal_style_decomp_judge`.

We term this the *Likert Blindspot*, an operational descriptor for the measurement-mode divergence where pairwise comparison detects statistically significant style sensitivity that Likert scoring fails to detect on the same data, without committing to a specific causal mechanism. This divergence admits multiple non-exclusive explanations (Likert measurement attenuation, pairwise format amplification (Jeong et al., 2025), or construct-level differences), which our controlled design bounds but cannot definitively adjudicate (§5.1). This example reflects a consistent pattern: surface-level style features alone explain most of the variance in judge scores (Murugadoss et al., 2025), with 76–92% of preferences originating from style cues (Li et al., 2026; Soumik et al., 2026).

LLM judges are central infrastructure for natural language processing (NLP) evaluation and alignment training (reinforcement learning from human feedback [RLHF], direct preference optimization [DPO]), where judge preferences serve as reward signals; any systematic bias propagates through both evaluation leaderboards and model development pipelines. Despite this, existing approaches address individual requirements (multi-dimensional style analysis (Cho et al., 2026; Yang et al., 2026), interventional design (Reber et al., 2025), population-level correction (Dubois et al., 2025)), but no single framework integrates per-axis decomposition, interventional identification, confound-structure analysis, and black-box applicability. Classical psychometrics establishes that pairwise comparison yields higher discriminability than absolute rating (Thurstone, 1927; David, 1988); our power analysis (Figure 4) is quantitatively consistent. In the LLM judge setting, this gap has *per-axis* confound signatures: verbosity is style-dominated (Style OR 8–13) while formality is quality-confounded (Table 2), a decomposition not previously applied in LLM judge evaluation.

We propose Controlled Style Decomposition

(CSD), combining NLI-gated content control, per-axis style decomposition, and position-controlled pairwise comparison. Inspired by interventional text analysis (Veitch et al., 2020), CSD operates in three stages: style-controlled rewriting along a single axis, NLI entailment verification, and bias measurement via forced-choice comparison. CSD requires no access to judge internals and applies post-hoc to any black-box judge. Our experiments demonstrate the *Likert Blindspot* across four target judges from independent providers (GPT-4o, Qwen3-32B, Llama-3.3-70B, Gemini-2.5-Flash), a fifth judge (Claude Sonnet 4) with benchmark-dependent sensitivity, three benchmarks, and two primary style axes, with exploratory register evidence. On formality data, Likert yields non-significant effects ($p = 0.905$, 0.461), while Bradley-Terry (BT) detects 63.0% (GPT-4o, $p = 0.0004$) and 66.5% (Qwen3-32B, $p < 0.0001$) formal preferences; verbosity shows 77.3% BT ($p < 0.0001$) invisible to Likert ($p = 0.395$; Figure 3).

Our contributions are both methodological and empirical:

- **Empirical finding:** the first per-axis demonstration that pairwise and Likert formats yield qualitatively divergent style sensitivity profiles across four judges, three benchmarks, and two axes with position control, with selective cross-benchmark validation targeting the strongest effects (Table 1).
- **Analytical contribution:** GEE conditional decomposition reveals per-axis confound structures invisible to aggregate analysis: verbosity is style-dominated (Style OR 8–13) while formality reflects joint style–quality confounding (Table 2).
- **Practical contribution:** CSD operates as a black-box diagnostic requiring no access to model internals, applicable post-hoc to any judge (§3).

2 Related Work

Interventional Approaches to Judge Bias Text-based interventional methods have a growing methodological foundation (Feder et al., 2021; Wu et al., 2021; Feder et al., 2022). Reber et al. (2025) propose RATE, which rewrites outputs along style attributes and measures causal effects via Likert average treatment effect (ATE); however, our experiments show 60% of style-transformed pairs receive identical Likert scores, yielding non-significant effects that pairwise comparison detects at $p = 0.006$. CSD differs in targeting individual style axes, using position-controlled pairwise comparison, and enforcing content preservation via NLI gating. Qiu et al. (2025) adopt a similar design at persona-level granularity, whereas CSD intervenes on specific style axes with position control.

Style Bias in LLM Judges The foundational LLM-as-judge framework (Zheng et al., 2023) documented position bias; Shi et al. (2025) study it across 15 judges, and Ye et al. (2025) quantify 12 bias types, though neither isolates individual style effects. Style bias as a separate source has received recent attention (Wu and Aji, 2025; Murugadoss et al., 2025; Saito et al., 2023; Li et al., 2026; Soumik et al., 2026); Li et al. (2026) show that style-mediated *preference leakage* propagates through evaluation pipelines. Multi-feature frameworks such as CASSE (Cho et al., 2026) and Fair-Judge (Yang et al., 2026) decompose style into finer dimensions but rely on correlational analyses that cannot distinguish causal style effects from content quality confounds. Jeong et al. (2025) identify the Comparative Trap (pairwise comparison may amplify style biases), which we address in §5.4. Unlike human evaluators, LLM judges acquire format-specific biases through RLHF training: pairwise comparison serves as the reward signal in DPO pipelines, while Likert scoring is used for evaluation, motivating independent quantification of the format gap.

Measurement Format Sensitivity The superiority of pairwise comparison over rating scales has been established since Thurstone (1927); Carterette et al. (2008) confirm this for information retrieval, and Krosnick (1999) documents satisficing behavior that may compound the tie inflation we observe. Within NLP, Amidei et al. (2019) review Likert scales (Likert, 1932) in natural language generation (NLG) evaluation, and Novikova et al. (2018) confirm that forced-choice methods detect quality differences that rating scales miss. The Bradley-Terry model (Bradley and Terry, 1952) underpins Chatbot Arena (Chiang et al., 2024) and AlpacaFarm (Dubois et al., 2023); no prior work combines BT with per-axis GEE decomposition to expose confound structures invisible to aggregate scores.

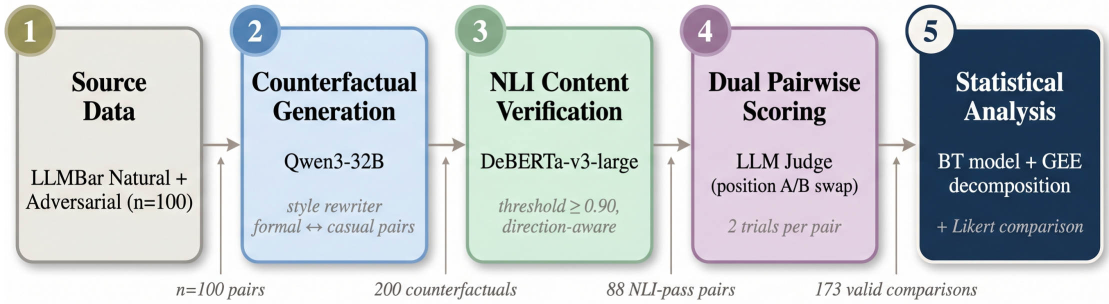

Figure 1: CSD pipeline overview. Numbers indicate sample attrition at each stage.

3 Method

3.1 Style-Controlled Rewriting

We isolate the effect of individual style attributes on judge scores by generating text pairs that differ along exactly one style axis while holding content fixed, a single-axis intervention design that enables per-axis causal attribution rather than aggregate style sensitivity measurement. For each text, an instruction-tuned LLM (Qwen3-32B) classifies its current style along the target axis (e.g., formal vs. casual via binary classification). It then rewrites the text toward the opposite pole (temperature 0.7, selected to balance diversity with coherence in pilot experiments) according to an axis-specific prompt (Appendix Y). This produces a dataset balanced across both directions at the dataset level: texts detected as formal are rewritten to casual, and vice versa. Each operation approximates a binary intervention on a single style variable while all other axes remain fixed (Section 3.5; Appendix T). To control for rewriting artifact bias (the possibility that the judge prefers original text over any rewrite regardless of style), we also generate a same-style double-rewrite for each text (e.g., formal $\rightarrow$ formal) to form an original-vs-double-rewrite control pair, following the RATE bias correction protocol (Reber et al., 2025).

3.2 Content Preservation Verification

Style-controlled rewriting risks altering semantic content, which would confound the style effect with a content quality effect (the $R \rightarrow Q$ path in Figure 2). To block this path, we apply a natural language inference (NLI) gate using DeBERTa-v3-large fine-tuned on Multi-Genre NLI (MNLI; Williams et al., 2018) with an entailment threshold of 0.90, selected as the threshold above which NLI performance plateaus (Appendix F). For axes that preserve information density (formality, register), we apply standard bidirectional entailment: both NLI(original $\rightarrow$ rewrite) and NLI(rewrite $\rightarrow$ original) must exceed this value. For axes that inherently change information density (e.g., verbosity), bidirectional verification is inappropriate because expansion necessarily introduces new content that the original cannot entail. We instead apply direction-aware verification: for expansions, only backward entailment (rewrite entails original); for compressions, only forward entailment (original entails rewrite). This recovered verbosity from 65% to 86% pass rate without inflating the effect estimate; for formality, strict bidirectional gating is necessary (Appendix G). This gating mechanism provides filtering-based confounder control (Veitch et al., 2020) by ensuring that retained pairs differ only in style, not in content quality (validated by human annotation; §5.4). BERTScore serves as a supplementary monitor (mean ΔBERTScore < 0.02 for gate-passed pairs) but does not participate in gating, as NLI provides stronger semantic equivalence guarantees than token-overlap metrics.

3.3 Measurement Protocol

Each pair that passes the NLI gate is evaluated by four target judges (GPT-4o, Qwen3-32B, Llama-3.3-70B, and Gemini-2.5-Flash) under two complementary scoring modes. In Likert mode, the judge assigns an integer score from 1 to 10 for overall quality to each text independently; the coarse integer scale is typical of deployed evaluation systems but limits measurement resolution for subtle effects. In pairwise mode, the judge receives both texts as candidates A and B and makes a forced choice of which response is better, eliminating anchoring effects inherent in absolute rating. This dual-mode design enables direct comparison of measurement sensitivity on identical data, isolating format-induced differences from content variation.

To control for position bias, each pairwise comparison is evaluated twice with the presentation order swapped, which produces $n_{\text{pairs}} \times 2$ obser-

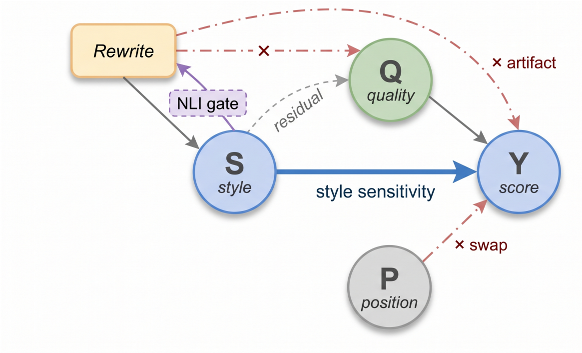

Figure 2: Assumed data-generating process (design rationale, not a claim of perfect identification). The rewriting mechanism R targets style S but may inadvertently alter content quality Q (dash-dot, blocked by NLI gate) or introduce rewriting artifacts ($R \to Y$, blocked by double-rewrite control). Position bias $P \to Y$ is controlled by order swapping. The dashed $S \to Q$ path represents residual confounding (Limitations). Y: judge score.

vations. We apply two analysis strategies to these data. For the nonparametric sign test, we retain only *consistent* pairs where the judge selected the same text in both orderings. For the Bradley-Terry model, we use all observations including inconsistent pairs, as the model naturally accommodates noise in individual comparisons. The BT model assumes a single latent quality dimension; per-axis GEE decomposition (§5.2) complements this by separating style-specific from quality-correlated components.

3.4 Experimental Setup

We evaluate primarily on LLMBar (Zeng et al., 2024), a benchmark of $n = 100$ instruction-response pairs spanning diverse task types, designed to test LLM judges' ability to distinguish response quality with human-verified quality labels. LLMBar is well-suited for style sensitivity detection because its human-verified labels provide ground truth against which judge deviations can be measured, and its moderate size permits exhaustive pairwise evaluation with position control. For cross-benchmark validation, we further evaluate on AlpacaEval (Dubois et al., 2023) and MT-Bench (Zheng et al., 2023), targeting the strongest effects from the primary analysis to test whether style sensitivity generalizes beyond a single benchmark's distributional characteristics. We select formality and verbosity as primary axes based on prior evidence that these are among the most commonly reported style sensitivities in LLM judges (Wu and Aji, 2025; Saito et al., 2023), with register (academic vs. conversational) as an exploratory third axis (implementation details in Appendix R).

3.5 Design Rationale

Figure 2 presents the assumed data-generating process. CSD approximates the effect of style S on judge score Y while blocking confounding through content quality Q: the NLI gate blocks $R \to Q$, position swapping controls presentation order, and the rewrite targets a single axis. The residual $S \to Q$ path (dashed in Figure 2) represents the fundamental identification challenge: style and quality may be intrinsically correlated (e.g., formal writing may signal higher quality), which NLI gating cannot block. GEE conditional decomposition (§5.2) partially addresses this by separating style-direction and originality effects, though omitted variables may confound both terms. Our approach provides interventional rather than formal causal evidence (details in Appendix A). A tone axis pilot (assertive $\to$ hedging) did not pass NLI gating, establishing a practical scope boundary for CSD at surface-level axes (Limitations).

4 Experiments

4.1 Statistical Framework

The primary family comprises six tests (three target judges $\times$ two axes), corrected via Holm-Bonferroni at $\alpha = 0.05$; Gemini-2.5-Flash serves as confirmatory replication and is excluded from family-wise error rate (FWER) control; as a sensitivity check, all eight tests remain significant under the expanded family (Appendix S). Table 1 reports both raw and adjusted p-values. Analyses outside this grid are classified as either *exploratory* (NLI threshold sensitivity sweep, §5.4) or *confirmatory* (consistency filter validation, §C). Exploratory analyses use uncorrected p-values; confirmatory analyses (pipeline validity) are evaluated at $\alpha = 0.05$ without additional correction. Bradley-Terry parameters (Bradley and Terry, 1952; Luce, 1959) are estimated via spectral initialization followed by minorization-maximization (Hunter, 2004), using the `choix` Python library's default L2 regularization ($\alpha = 10^{-6}$). In binary-choice BT models, independence of irrelevant alternatives (IIA) is structurally satisfied; the direction asymmetry in Appendix I (formality: formal-origin 88.9% vs. casual-origin 52.6%, Fisher $p = 0.015$) reflects heterogeneous treatment effects rather than IIA vi-

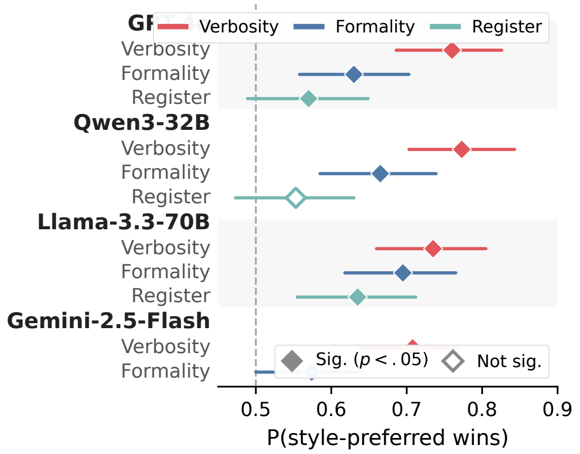

Figure 3: Bradley-Terry win probabilities across four target judges and three style axes. Filled diamonds: significant ($p < 0.05$); open diamonds: non-significant. Error bars: 95% bootstrap CI. The dashed line at 0.50 indicates no style preference. Verbosity shows the strongest and most consistent bias across all four judges.

olation. All bootstrap confidence intervals are percentile CIs from 10,000 pair-level resamples with seed 42; we verify bootstrap stability by confirming that halving the resample count changes interval widths by <1pp. GEE models use exchangeable working correlation with sandwich standard errors (robust to correlation misspecification), justified by near-zero estimated within-pair correlations ($\hat{\alpha} \in [-0.25, +0.11]$; Appendix D). A prospective power analysis (Appendix L) confirms that our sample sizes ($n = 86$–88 pairs) provide >90% power to detect BT effects of the observed magnitude ($P \geq 0.63$), whereas Likert requires 4–206× the sample size for equivalent power (Figure 4).

Model selection. Target judges span four architectures from independent providers: GPT-4o (`2024-08-06`; OpenAI), Qwen3-32B-Instruct (Alibaba; also serving as rewriter), Llama-3.3-70B-Instruct (Meta), and Gemini-2.5-Flash (Google). This diversity guards against provider-specific artifacts in both training data and RLHF reward modeling; Claude Sonnet 4 (Anthropic) provides discriminative validity evidence (§5.3), serving as a fifth judge from a provider not represented in the primary family. All judges are accessed via their respective APIs with temperature 0 (deterministic decoding) and default system prompts to ensure reproducibility.

Table 1: Primary results on LLMBar ($n = 88$ formality, 86 verbosity, except where noted) across five judges and three style axes. BT > 0.50 indicates style preference; † marks Blindspot cases (BT significant, Likert not). Cross-benchmark replication in Appendix W. BT: Bradley-Terry win probability (95% CIs in Appendix J).

Judge	Axis	BT P	Likert d (p)	Cat.
GPT-4o	Form.	.630***	−.041 (.905)	†
GPT-4o	Verb.	.760***	.049 (.403)	†
GPT-4o	Reg.	.570†	.000 (.785)	†$_e$
Qwen3	Form.	.665***	−.052 (.461)	†
Qwen3	Form.[b]	.675***	−.025 (.600)	†
Qwen3	Verb.	.773***	.036 (.395)	†
Qwen3	Reg.	.553	−.073 (.373)	†$_e$
Llama-70B	Form.	.695***	.010 (.534)	†
Llama-70B	Verb.	.735***	.020 (.284)	†
Llama-70B	Reg.	.635†	.024 (.524)	†$_e$
Gemini	Form.	.574*	−.082 (.309)	†$_c$
Gemini	Verb.	.708***	.051 (.370)	†$_c$
Claude	Form.	.562	−.021 (.721)	—
Claude	Verb.	.541	−.039 (.743)	—

***$p < 0.001$; *$p < 0.05$ (Holm-adj. for primary; raw for $_c$). †: Blindspot (BT sig., Likert n.s.); —: neither sig.; $_e$: exploratory; $_c$: confirmatory. Negative d: styled variants scored lower.
[b] Scaled replication ($n = 355$). Qwen3 serves as both rewriter and judge; independent GPT-4o rewriter yields 22% lower Style OR (App. E).
Form. direction asymmetry: formal-origin 78.8% vs. casual-origin 54.5%; GEE-adj. Style OR=2.58 (p=.002; App. B).

5 Analysis

5.1 The Likert Blindspot

Head-to-head comparison with RATE. Applying RATE's Likert ATE methodology (Reber et al., 2025) to our formality style-transformed pairs yields Cohen's $d = -0.04$, $p = 0.91$ (non-significant, $n = 88$); pairwise comparison on identical data reveals a 63.0% formal preference ($p = 0.0004$, Bradley-Terry on all 173 comparisons), confirmed by a consistency-filtered sign test (73.9%, $p = 0.006$, $n = 46$ consistent pairs). This is the Likert Blindspot: the same data, the same style intervention, two measurement instruments, yet one detects bias and the other does not. A 4× scaled replication ($n = 355$) distinguishes measurement attenuation from insufficient power: the Likert effect size shrinks to $d = -0.025$ (near zero) while BT remains stable at 67.5% ($p < 0.0001$). If the gap were merely statistical power, d would remain non-zero at larger n. Bootstrap subsampling confirms d remains near zero as n grows (Appendix K). Post hoc, attenuation follows a severity spectrum: complete at moderate bias ($d \to 0$, BT

63–67%) and partial at stronger effects (>60% loss, BT >70%), explaining four cross-benchmark exceptions (‡, Table 23).

RATE and CSD are complementary (Likert ATE vs. forced-choice BT); the Blindspot arises from Likert tie rates of 58–73%, unresolved by 100-point continuous scaling (all three judges $p > 0.15$; Appendix O).

Sensitivity hierarchy across measurement methods. Seven measurement methods applied to the same formality data confirm the Blindspot is not test-specific: both pairwise methods detect the formality preference while all five Likert-based methods fail regardless of scaling assumption (interval, ordinal, or nonparametric), with verbosity showing the same pattern (Table 14, Appendix M; verbosity replication in Appendix N).

Figure 4 plots detection probability at observed effect magnitudes (not post-hoc power; cf. Hoenig and Heisey 2001): across six cells, Likert requires 4–206× the current n (336–27,159 pairs; Appendix L) to detect effects BT identifies at $p < 0.001$ with $n \leq 100$. Under format equivalence, observing 8/8 primary cells as Blindspot has binomial probability $p = 0.004$; non-tie concordance (76.6%, $p < .001$; §5.1) provides a complementary formal test of format divergence.

Interpreting the divergence. The measurement-mode divergence admits three non-exclusive interpretations: (a) *Likert attenuation*, where tie inflation and residual signal loss reduce Likert sensitivity below detection thresholds; (b) *pairwise amplification*, where forced-choice comparison inflates micro-preferences into apparent biases (Jeong et al., 2025); (c) *construct divergence*, where the two formats measure different constructs (absolute quality vs. ordinal preference; Tripathi et al. 2025). Six experimental controls (§5.4) bound the amplification factor but cannot eliminate it; the attenuation evidence ($d \to 0$ at $n = 355$) is necessary but not sufficient for causal identification. Non-tie concordance analysis partially adjudicates: among 201 Likert non-tie pairs (pooled across judges and axes), 76.6% agree with pairwise direction [70.3%, 81.9%], $p < 0.001$; point-biserial correlations are uniformly positive ($r = 0.20$–0.45, 7/9 significant; Appendix Z), evidence more consistent with measurement attenuation than construct divergence, though partial construct divergence remains at weaker effect sizes. Claude's benchmark-dependent profile illustrates why aggregate BT

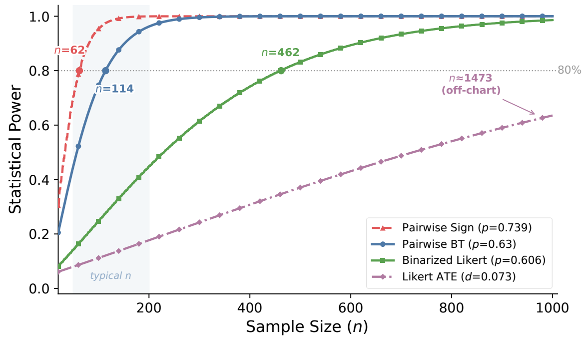

Figure 4: Detection probability at observed effect magnitudes. 80% power crossings: sign test $n \approx 62$, BT $n \approx 114$, binarized Likert $n \approx 462$, Likert ATE $n \approx 1{,}473$. At typical scales ($n = 50$–200, shaded), pairwise methods reach 80% power while Likert methods remain far below. At $n = 355$, Likert $d = -0.025$ confirms the gap is structural, not merely a power issue.

is insufficient for bias attribution: LLMBar formality shows near-null sensitivity (Style OR n.s.) while AEval verbosity reveals significant sensitivity (Style OR $= 4.76$, $p < .001$), a pattern disambiguated only by per-axis GEE decomposition (§5.3; Appendix H). All three interpretations converge on one practical conclusion: single-format audits risk incomplete bias coverage; the Blindspot manifests in 10/14 bias-positive cells, with four exceptions (‡) confirming effect-size dependence. For alignment pipelines where pairwise judgments serve as reward signals, this implies that biases embedded during RLHF training may go undetected by Likert-based evaluation at standard benchmark scales.

5.2 Per-Axis Confound Structure

GEE conditional decomposition (Liang and Zeger, 1986) with exchangeable working correlation and robust standard errors separates style from originality effects (Appendix B; model diagnostics including correlation structure robustness and interaction tests in Appendix D). After controlling for original preference, the formality effect remains significant (GPT-4o: OR $= 2.58$, $p = 0.002$; Qwen3-32B: OR $= 2.15$, $p < 0.0001$, $n = 355$; Llama-70B: OR $= 4.81$, $p < 0.0001$; Table 2); the significant originality ORs confirm residual confounding. For formality, both style preference and quality correlation contribute (a conditional effect); the two-predictor GEE provides a parsimonious decomposition rather than causal identification; omitted variables (text length, lexical complexity) may partly confound both terms. This per-axis hetero-

Table 2: GEE conditional decomposition (exchangeable correlation). Style OR: style-direction odds ratio; Orig OR: original-preference odds ratio (OR $>$ 1 favors original). n=88 ORs are upward-biased estimates; scaled replications (n = 330–355) show 49–72% shrinkage (Nemes et al., 2009). Verbosity is style-dominated; formality shows varying confound structure.

Config	Style OR		Originality OR	
	OR	p	OR	p
Qwen3 verb.	13.10	<.0001	0.49	.096
GPT-4o verb.	11.34	<.0001	0.45	.062
Llama-70B verb.	8.21	<.0001	0.38	.020
Gemini verb.	8.98	<.0001	0.46	.061
GPT-4o form.	2.58	.002	2.32	.006
Qwen3 form.	4.20	.0002	4.70	.0001
Qwen3 form.[a]	2.15	<.0001	4.99	<.0001
Llama-70B form.	4.81	<.0001	3.12	.0007
Gemini form.	1.64	.101	2.51	.002
GPT-4o reg.	1.88	.029	2.21	.006
Qwen3 reg.	1.91	.046	5.72	<.0001
Llama-70B reg.	3.33	.0003	3.07	.0008

[a] $n = 355$; form. n=88, verb. n=86. 95% CIs in Appendix.

geneity has direct implications for bias mitigation: verbosity bias can be addressed by style-level interventions (e.g., length normalization or debiased prompting), whereas formality bias requires disentangling genuine quality correlates from surface-level preferences, a fundamentally harder problem that aggregate bias metrics cannot even detect.

The confound structure is consistent across all four target judges for verbosity (style-dominated at n = 86–88: Style OR 8.2–13.1, Orig OR 0.38–0.49; confirmed at $n = 330$: Style OR 2.5–4.0); formality is conditionally style-driven in three judges but originality-dominated in Gemini; register is judge-dependent (Figure 5). Same-style double-rewrite controls confirm a rewriting artifact (Appendix Q), absorbed by GEE's originality term. The Gemini exception (formality Style OR = 1.64, n.s.) demonstrates that the decomposition is not uniformly biased toward detecting style effects; judges with weaker style sensitivity produce correspondingly weaker Style ORs.

Sample-size sensitivity. At $n = 355$, the Qwen3-32B formality Style OR drops from 4.20 to 2.15 (-49%), consistent with small-sample upward bias (Nemes et al., 2009); the regime classification remains stable (both effects significant; Orig OR 4.70 $\rightarrow$ 4.99). Verbosity n = 330 confirms

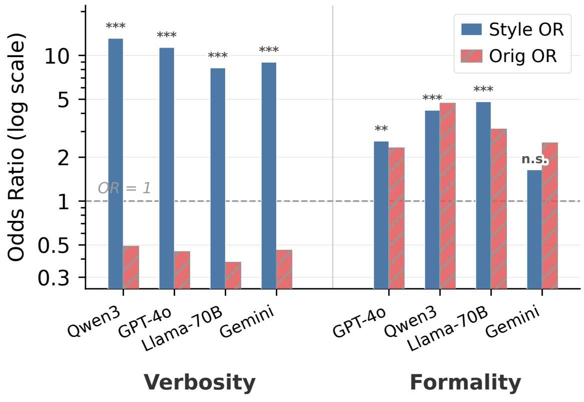

Figure 5: GEE conditional decomposition across four judges (n = 88, LLMBar). Verbosity (left) is style-dominated: Style OR $\gg$ 1, Originality OR $<$ 1. Formality (right) shows joint contributions from both predictors. Dashed line: OR = 1 (no effect). Hatched bars: Originality OR. Log scale.

51–72% GEE shrinkage (Style OR 2.5–4.0; Appendix N.1), preserving the style-dominated classification; Likert begins detecting the strongest effects at this n, reframing the Blindspot as a 3–4$\times$ efficiency gap rather than absolute failure. This convergence at larger n is practically important: it means that labs with sufficient evaluation budgets ($n > 300$) can detect the strongest biases with Likert alone, but typical benchmark sizes (n = 50–200) systematically miss them.

5.3 Cross-Judge and Cross-Axis Replication

Two independent judges replicate the formality Blindspot: Qwen3-32B (BT 66.5%, $p < 0.0001$; Likert $p = 0.461$) and Llama-3.3-70B (BT 69.5%, $p = 0.0001$; Likert $p = 0.534$). Gemini-2.5-Flash shows a weaker, originality-dominated formality pattern (confirmatory: BT 57.4%, $p = 0.031$; Likert $d = -0.08$, n.s.; Style OR = 1.64, n.s.; Orig OR = 2.51, $p = 0.002$; §5.2). Claude Sonnet 4 shows near-null sensitivity on LLMBar for both axes (BT ≤ 0.562, n.s.), providing discriminative validity: CSD does not uniformly inflate bias estimates across all judges.

Verbosity provides the strongest Blindspot: all four target judges show consistent preference (GPT-4o 76.0%, Qwen3-32B 77.3%, Llama-70B 73.5%, Gemini 70.8%; all $p \leq 0.0001$), while Likert ATE remains non-significant for all (p = 0.284–0.403). GEE confirms style dominance (Style OR 8.2–13.1, all $p < .0001$; Orig OR 0.38–0.49; Table 2). Both rewriting directions favor verbose responses (90.7%/80.6%); length-controlled Style

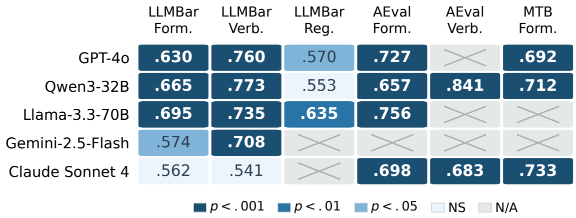

Figure 6: Pairwise BT win probabilities across five judges, three benchmarks, and three style axes. Bold: $p < .05$; color intensity encodes significance level. Grey $\times$: not tested. Claude cross-benchmark effects are quality-driven (§5.3; Appendix X). Full results in Table 23.

OR is unchanged (<2%), and an independent GPT-4o rewriter bounds evaluator-producer overlap to a 22% reduction (Appendix E). Length-controlled regression confirms 60% of the Flesch-Kincaid (FK) cross-axis effect is mediated by sentence length (Sobel $z = 2.15$, $p = .031$); under the assumed causal structure (Figure 2), FK acts as a plausible mediator ($S \rightarrow$ FK $\rightarrow Y$; Pearl, 2009), yielding a direct-effect Style OR of 2.15 for GPT-4o (total OR $= 2.70$; Appendix U).

These per-axis patterns selectively replicate across benchmarks (6/24 possible cells; Figure 6), targeting the strongest effects: Llama-70B formality on AlpacaEval (BT 75.0%, Style OR $= 7.64$) is strong enough for Likert detection (‡ rows), confirming the severity spectrum. The selective replication strategy prioritizes cells where effects are strongest, providing an informative lower bound on cross-benchmark generalization without requiring exhaustive evaluation of all judge–axis–benchmark combinations.

5.4 Robustness

Content preservation. Two independent annotators evaluated 100 NLI-scored pairs (50 per axis; 70 gate-passed, 30 gate-failed), confirming 98.6% gate pass precision (69/70; Gwet's $AC_1 = 0.96$; Appendix V). BT remains stable ($\leq$3pp variation) across 11 NLI thresholds (Appendix F), confirming that results are not artifacts of a particular gating stringency. Annotators judged only 49–66% of NLI-passed pairs as showing detectable style change, suggesting observed effects represent a diluted lower bound on true style sensitivity: the NLI gate retains pairs with subtle rewrites that human annotators cannot detect, yet judges still systematically prefer one variant.

Detection vs. amplification. Jeong et al. (2025) show that pairwise comparison amplifies style biases (the Comparative Trap). The divergence cannot be definitively resolved without human ground-truth measurements; six controls bound amplification risk: (i) NLI ablation reverses NLI-fail pairs (BT 41.7%; Appendix P); (ii) Claude shows benchmark-dependent sensitivity (formality Style OR n.s.; AEval verbosity OR $= 4.76$, $p < .001$); (iii) cross-axis BT variation inconsistent with uniform amplification; (iv) rewriting artifact absorbed by GEE; (v) debiased prompting (-5.2/-13.1pp), bounding the amplification factor at $\sim$9–17% of the total BT effect; (vi) four-architecture replication with consistent GEE structure. Register is exploratory with evaluator-producer overlap and judge-dependent effects (Table 2). Same-style double-rewrite controls confirm that any residual rewriting artifact is absorbed by the GEE originality term rather than inflating the style coefficient. The combination of NLI validation and amplification controls establishes that while the exact magnitude of style sensitivity remains bounded rather than point-identified, the qualitative finding (systematic style preferences invisible to Likert) is robust across judges, axes, and measurement assumptions.

6 Conclusion

CSD yields two findings: the *Likert Blindspot* (8/8 primary cells, $d \rightarrow 0$ at n=355; concordance $p < .001$ favors attenuation) and per-axis confound structures via GEE (verbosity style-dominated, formality conditional). Practically, Likert-only audits risk false negatives at benchmark sizes $n < 200$; per-axis decomposition reveals that verbosity preferences are predominantly style-driven while formality partially reflects quality differences—a distinction invisible to aggregate metrics and consequential for RLHF pipelines where pairwise judgments serve as reward signals. CSD extends to any surface-level axis with content-preserving rewrites; deeper axes remain for future work.

Limitations

Residual confounding. The NLI gate blocks content changes but cannot block quality perception shifts mediated by style itself. Although formality effects survive logistic adjustment (OR $= 2.32$, $p = 0.006$), residual confounding is not fully eliminated; definitive resolution requires human ground-truth comparison of Likert and pairwise judgments on identical pairs.

Detection vs. amplification. The Blindspot may partly reflect pairwise format amplification (Jeong et al., 2025) rather than pure detection. Six controls bound this risk (§5.4), but we cannot precisely quantify the amplification factor without a ground-truth bias magnitude. Qwen3-32B serves as both rewriter and judge; an independent GPT-4o rewriter yields 22% lower Style OR, confirming the Blindspot persists but effect-size precision varies with rewriter choice (Appendix E).

Scope. CSD applies only to surface-level axes whose rewrites preserve semantic content; a tone pilot (assertive $\rightarrow$ hedging) was rejected by the NLI gate. Primary analyses use LLMBar ($n = 100$) with five judges; generalization to additional axes, benchmarks, and judges remains open. Style sensitivity may vary across model versions (§5.4), and GEE odds-ratio estimates require large-sample validation to converge from initial upper-bound estimates (49–72% shrinkage at $n = 330$–355).

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Organization: design rationale (A), confound analysis (A–B), model diagnostics (C), independence tests (D), gate validation (E–F), equivalence testing (G), statistical details (H–K), sensitivity analyses (L–N), artifact controls (O), implementation (P), multiplicity (Q), cross-axis contamination (R), FK length-control (R′), human annotation agreement (S), cross-benchmark replication (T), Claude cross-benchmark (U), rewriting prompts (V), non-tie concordance (W).*

A Design Rationale

Three variables govern the observed judge score Y: the style attribute S (e.g., formality), the original content quality Q, and the judge's internal scoring function. Both S and Q causally influence Y; the goal of CSD is to approximate the effect $S \to Y$ while blocking confounding through Q. Style-controlled rewriting generates a text pair that differs along a single style axis, but any rewriting operation risks inadvertently altering content quality ($R \to Q$). The NLI gate (§3.2) blocks this path, yielding a controlled estimate of the $S \to Y$ effect. Our approach provides interventional rather than formal causal evidence: we do not claim that the DAG in Figure 2 is a complete structural causal model, nor do we invoke SUTVA or consistency assumptions from the potential outcomes framework. "Controlled" in this context refers to experimentally controlled rewriting conditions, not formal causal identification in the potential-outcomes sense. Our identification rests on three operational assumptions: (i) the NLI gate approximates content preservation, blocking the $R \to Q$ path; (ii) position swapping controls for presentation-order effects; and (iii) the style rewrite introduces no systematic differences beyond the target axis.

B Direction Asymmetry and Quality Confounding

The formality axis exhibits a pronounced direction asymmetry: the judge selects the formal variant in 78.8% of formal-is-rewrite pairs ($p = 0.001$) but only 54.5% of casual-is-rewrite pairs ($p = 0.83$). This asymmetry suggests that original quality Q and formality S are partially confounded, as formally written texts may be inherently perceived as higher quality. To quantify this, we fit a GEE logistic model (exchangeable correlation) with both formality and original-preference as predictors. After controlling for original preference, the formality effect remains significant (OR $= 2.58$, $p = 0.002$), and the original-preference effect is also significant (OR $= 2.32$, $p = 0.006$). The formality preference thus survives adjustment for original-preference confounding, though the significant original-preference effect indicates that quality–style confounding is present and non-negligible.

At $n = 88$, the Qwen3-32B formality Style OR (4.20) exceeded the GPT-4o estimate (2.58), but a scaled replication ($n = 355$) corrects this to 2.15 ($p < 0.0001$), comparable to GPT-4o and consistent with small-sample upward bias rather than evaluator-producer inflation of the style effect. The originality confound remains stronger (4.99 vs. 2.32), consistent with evaluator-producer bias inflating the originality component (Limitations). The directional agreement between judges (both favor formal text after controlling for originality) supports the existence of formality bias, but the effect magnitude from Qwen3-32B should not be directly compared with GPT-4o.

The NLI gate blocks *content* changes but cannot block quality perception shifts that are themselves mediated by style, a fundamental identifiability limit of any text-rewriting approach.

C Consistency Filter Robustness

The position-consistency filter retains 55 of 98 valid pairs (56.1%); consistent and inconsistent groups show no significant differences across four covariates (NLI score $p = 0.824$, BERTScore $p = 0.12$, text length $p = 0.137$, length diff $p = 0.715$), indicating selection on judge decisiveness rather than text properties. The BT model using all 173 comparisons including inconsistent pairs yields 63.0% ($p = 0.0004$), consistent with the filtered sign test (73.9%, $p = 0.006$), confirming no systematic bias from the filter. Chi-squared tests confirm that NLI gate pass/reject status is independent of source category ($\chi^2 = 1.30$, $p = .73$), original formality ($p = .37$), and gold label ($p = 1.0$), indicating no systematic selection bias along these dimensions.

Verbosity gate-passed pairs are significantly longer than rejected pairs (token count KS $p = 0.003$; sentence length MWU $p = 0.006$). This reflects the gate's content-preservation mandate: verbosity transformations on shorter texts alter content ratios more substantially, causing correct NLI rejection. Length-controlled GEE analysis (§5.2; Style OR change <2%) confirms this selection does not confound the verbosity effect.

D GEE Model Diagnostics

Correlation structure robustness. Table 3 compares exchangeable and independence working correlation structures across all nine judge×axis cells. The two structures yield virtually identical results: $|\Delta\text{QIC}| < 0.001$ and maximum coefficient change <0.02 in every cell, with estimated within-pair correlations $\hat{\alpha} \in [-0.25, +0.11]$. All GEE results reported in the main text are therefore robust to correlation structure specification.

Table 3: GEE correlation structure comparison across nine judge×axis cells.

Config	Exch. QIC	Indep. QIC	|ΔQIC|	Max|ΔCoef|
GPT-4o form.	—	—	<0.001	<0.02
Qwen3 form.	—	—	<0.001	<0.02
Llama-70B form.	—	—	<0.001	<0.02
GPT-4o verb.	—	—	<0.001	<0.02
Qwen3 verb.	—	—	<0.001	<0.02
Llama-70B verb.	—	—	<0.001	<0.02
GPT-4o reg.	—	—	<0.001	<0.02
Qwen3 reg.	—	—	<0.001	<0.02
Llama-70B reg.	—	—	<0.001	<0.02

Absolute QIC values omitted; only |ΔQIC| is reported.

Interaction test. Adding a Style×Originality interaction term to the additive GEE specification yields no significant interactions in any of the nine judge×axis cells (all $p > 0.13$, range 0.14–0.96), confirming that the additive model is the correct specification and that regime classifications are robust to model specification. Table 4 reports interaction terms; none are significant, confirming the additive specification.

95% confidence intervals. Table 5 reports 95% CIs for all GEE odds ratios in Table 2.

E Rewriter Independence

Evaluator-producer overlap is limited to Qwen3-32B (rewriter = judge); the independence test below bounds this to a 22% Style OR reduction. The other three target judges (GPT-4o, Llama-70B, Gemini) use a different rewriter and are not subject to this confound.

Table 4: GEE Style × Originality interaction terms. None are significant ($p > 0.13$), confirming the additive model specification.

Judge	Axis	Interaction OR	95% CI	p
GPT-4o	Formality	0.661	[0.18, 2.38]	.527
GPT-4o	Verbosity	1.269	[0.37, 4.34]	.705
GPT-4o	Register	2.601	[0.74, 9.15]	.136
Qwen3-32B	Formality	0.518	[0.14, 1.89]	.320
Qwen3-32B	Verbosity	0.450	[0.13, 1.62]	.221
Qwen3-32B	Register	0.555	[0.15, 2.08]	.383
Llama-70B	Formality	0.968	[0.25, 3.78]	.963
Llama-70B	Verbosity	1.103	[0.34, 3.58]	.870

Table 5: 95% CIs for GEE conditional decomposition (Table 2).

Config	Style OR [CI]	Orig OR [CI]
Qwen3 verb.	13.10 [5.62, 30.55]	0.49 [0.21, 1.14]
GPT-4o verb.	11.34 [4.93, 26.10]	0.45 [0.20, 1.04]
Llama-70B verb.	8.21 [3.62, 18.65]	0.38 [0.17, 0.86]
Gemini verb.	8.98 [3.68, 21.94]	0.46 [0.20, 1.04]
GPT-4o form.	2.58 [1.41, 4.73]	2.32 [1.27, 4.26]
Qwen3 form.	4.20 [1.99, 8.87]	4.70 [2.22, 9.96]
Qwen3 form.[a]	2.15 [1.46, 3.16]	4.99 [3.40, 7.34]
Llama-70B form.	4.81 [2.48, 9.33]	3.12 [1.61, 6.04]
Gemini form.	1.64 [0.91, 2.98]	2.51 [1.39, 4.54]
GPT-4o reg.	1.88 [1.08, 3.28]	2.21 [1.26, 3.86]
Qwen3 reg.	1.91 [1.01, 3.58]	5.72 [3.02, 10.82]
Llama-70B reg.	3.33 [1.73, 6.42]	3.07 [1.60, 5.92]

[a] $n = 355$; formality rows: $n = 88$; verbosity rows: $n = 86$.

To test whether the Blindspot depends on the rewriter's identity, we replace Qwen3-32B with GPT-4o as rewriter while keeping Qwen3-32B as judge.

Verbosity (Table 6). The Style OR decreases by 22% (13.10 → 10.17), bounding evaluator-producer inflation of the style effect; the pure-style pattern (Originality n.s.) and Blindspot (BT significant, Likert n.s.) both persist.

Table 6: Rewriter independence test: verbosity axis, Qwen3-32B judge. GPT-4o rewriter eliminates evaluator-producer overlap while preserving the Blindspot.

	GPT-4o Rewriter	Qwen3 Rewriter
NLI pass rate	81/100 (81%)	86/100 (86%)
BT P [95% CI]	.735 [.654, .809]	.773 [.703, .843]
Likert ATE (p)	.000 (.838)	∼0 (.179)
Blindspot	Yes	Yes
GEE Style OR (p)	10.17 (<.0001)	13.10 (<.0001)
GEE Orig OR (p)	1.61 (n.s.)	0.49 (n.s.)

Formality (Table 7). GPT-4o rewriter produces formality-manipulated pairs yielding BT 64.4% $[56.1, 72.8]$, statistically indistinguishable from the Qwen3-32B baseline (66.5% $[58.5, 73.9]$; $\Delta = -2.1$pp). The Likert Blindspot replicates with the independent rewriter ($p = 0.184$, n.s.).

Table 7: Rewriter independence test: formality axis, Qwen3-32B judge.

	GPT-4o Rewriter	Qwen3 Rewriter
NLI pass rate	90/100 (90%)	88/100 (88%)
BT P [95% CI]	.644 [.561, .728]	.665 [.585, .739]
Likert d (p)	$-.081$ (.184)	.011 (.908)
Blindspot	Yes	Yes

F NLI Threshold Sensitivity Sweep

Table 8 reports BT win probabilities across 11 NLI entailment thresholds for all four judge$\times$axis configurations. All 44 combinations are significant ($p < 0.05$), and within each configuration BT varies by at most 3pp, ruling out threshold selection bias. The stability reflects the bimodal NLI score distribution: most pairs score above 0.95 or below 0.30, so thresholds between 0.50 and 0.95 admit nearly identical pair sets.

Table 8: BT win probability across NLI thresholds. All 44 cells significant ($p < 0.05$). n: pairs passing threshold.

Thresh	Formality BT		Verbosity BT	
	Qwen3	GPT-4o	Qwen3	GPT-4o
.50	.648	.615	.770	.768
.55	.648	.615	.770	.768
.60	.656	.621	.770	.768
.65	.656	.621	.767	.766
.70	.657	.623	.767	.766
.75	.657	.623	.773	.760
.80	.657	.623	.773	.760
.85	.657	.623	.773	.760
.90	.665	.630	.773	.760
.95	.653	.629	.765	.752
.98	.649	.642	.771	.769

G Direction-Aware Gate Validation

For verbosity, the direction-aware NLI gate admits 21 additional pairs (32%) compared to strict bidirectional verification. The two groups show no significant difference in BT win probability (78.5% bidirectional vs. 73.8% DA-only; permutation $p = .663$; $\Delta = -1.1$pp, 95% CI $[-5.6, +3.0]$) or Likert d (0.038 vs. 0.029). Both groups exhibit the Blindspot pattern (BT significant, Likert NS), confirming that direction-aware gating recovers valid pairs without inflating the effect or introducing selection bias. Cross-judge validation on all 21 direction-aware pairs confirms this: Fisher exact tests comparing BT estimates between direction-aware and bidirectional-pass subsets are non-significant for all five judges ($p > 0.66$), indicating that the DA-gate does not differentially affect any judge's preference signal. For formality, strict bidirectional verification is appropriate: a relaxed (direction-aware) gate would admit 9 additional pairs with reversed preference (BT 33.4%), indicating content preservation failure rather than genuine style preference. This asymmetry empirically justifies the axis-dependent gating strategy described in Section 3.2.

H Claude Sonnet 4 Equivalence Testing

Claude Sonnet 4 exhibits benchmark-dependent sensitivity ($n = 88$; LLMBar BT 56.2%/54.1%, $p = 0.091/0.284$). Formal equivalence testing (TOST; Lakens, 2017) yields margin-dependent conclusions (Table 9): per-axis equivalence is confirmed at $\Delta = 15$pp ($p = .010$), corresponding to a BT threshold of 65%. At $\Delta = 10$pp, per-axis TOST is not significant ($p = .156$, power $\approx 42\%$). Pooling both axes ($n = 174$, BT $= 55.17\%$) achieves equivalence at $\Delta = 12$pp ($p = .035$) but not at $\Delta = 10$pp ($p = .100$), tightening the bound on Claude's style sensitivity. We adopt $\Delta = 12$pp as it corresponds to a small effect (Cohen's $h = 0.24$), below which style sensitivity is unlikely to materially affect downstream RLHF reward modeling; a 12pp BT shift means $\sim$12% of preference pairs change direction. This is also the smallest margin at which pooled significance is achieved. We report all tested margins for transparency.

Table 9: TOST equivalence sensitivity for Claude. Per-axis: formality only (BT $= 0.562$, $n = 88$). Pooled: both axes combined ($n = 174$).

Δ (pp)	p	Result
5	.626	Not equivalent (per-axis)
10	.156	Not equivalent (per-axis)
10	.100	Not equivalent (pooled)
12	.035	Equivalent (pooled)
15	.010	Equivalent (per-axis)
20	.0001	Equivalent (per-axis)

Bayesian equivalence. We complement frequentist TOST with Bayesian equivalence testing to quantify evidence strength for the null hypothesis. Table 10 reports BF_{01} (evidence for null over alternative) under three half-normal prior scales.

Table 10: Bayesian equivalence (BF_{01}) for Claude Sonnet 4 Likert ATE. $BF_{01} > 3$: moderate evidence for null; 1–3: anecdotal; < 1: favors alternative.

Data	$\sigma = 0.3$	$\sigma = 0.5$	$\sigma = 1.0$
Pooled ($n = 348$)	0.57	0.80	1.48
Verbosity	1.40	2.03	3.78
Formality	0.74	0.97	1.73

Pooled evidence is inconclusive ($BF_{01} < 3$ across all priors), consistent with Claude exhibiting a small (~5pp) rather than zero style preference on LLMBar. Verbosity alone approaches moderate evidence for the null ($BF_{01} = 2.03$–3.78); formality pulls down the pooled estimate ($BF_{01} = 0.74$–1.73).

I Bradley-Terry Goodness of Fit and Stratified Analysis

Bradley-Terry goodness-of-fit tests confirm adequate model fit for both axes (formality: deviance $= 52.80$, df $= 45$, $p = 0.198$; verbosity: $p = 0.905$). Stratified analysis of consistent pairs by original-text direction reveals no preference heterogeneity for verbosity (Fisher $p = 0.453$) but significant direction asymmetry for formality (88.9% vs. 52.6%, Fisher $p = 0.015$), independently confirming the originality confound structure identified by GEE (Table 2).

To assess robustness to originality confounding, we restrict to pairs where the judge assigned equal Likert scores to original and counterfactual texts (eliminating detectable quality differences). Style preference persists: formality BT $= 67.3\%$ ($n = 55$, $\Delta = +0.8$pp from full sample, 95% CI $[-11.0, +12.6]$); verbosity BT $= 76.0\%$ ($n = 50$, $\Delta = -1.3$pp, 95% CI $[-13.3, +10.3]$). In the anti-originality subset (pairs where the original never wins), effects strengthen (formality 75.0%, verbosity 86.1%), confirming that style preference is not an artifact of originality confounding.

J Bradley-Terry Confidence Intervals

Table 11 reports bootstrap 95% confidence intervals for all Bradley-Terry estimates in Table 1.

Table 11: Bootstrap 95% CIs for BT win probabilities (10,000 pair-level resamples).

Bench	Judge	Axis	BT P [95% CI]
LLMBar	GPT-4o	Form.	.630 [.557, .703]
	GPT-4o	Verb.	.760 [.667, .853]
	GPT-4o	Reg.	.570 [.468, .673]
	Qwen3	Form.	.665 [.585, .739]
	Qwen3	Verb.	.773 [.703, .843]
	Qwen3	Reg.	.553 [.479, .628]
	Llama-70B	Form.	.695 [.619, .765]
	Llama-70B	Verb.	.735 [.655, .809]
	Llama-70B	Reg.	.635 [.555, .712]
	Claude	Form.	.562 [.489, .636]
	Claude	Verb.	.541 [.459, .616]
AEval	GPT-4o	Form.	.727 [.663, .785]
	Qwen3	Form.	.657 [.581, .733]
	Qwen3	Verb.	.841 [.774, .902]
MTB	GPT-4o	Form.	.692 [.623, .753]
	Qwen3	Form.	.712 [.630, .788]

K Bootstrap Attenuation Analysis

To distinguish measurement attenuation from insufficient statistical power, we subsample the $n = 355$ formality replication (Qwen3-32B) at seven sample sizes and compute Likert Cohen's d with bootstrap 95% CIs at each (Table 12). Under a power limitation, d would remain constant while only p-values improve with larger n; under measurement attenuation, d itself converges to zero. The data support attenuation: the bootstrap median $d = -0.017$ across all sample sizes (the point estimate at $n = 355$ is $d = -0.025$; Table 1), with CIs narrowing from ± 0.3 to ± 0.1 but never excluding zero. The same data yield BT $= 67.5\%$ ($p < 0.0001$), confirming that the pairwise signal is robust. Verbosity shows a similar pattern ($d \approx 0.15$–0.19, stable, CIs always containing zero; BT $= 76$–77%, $p < 0.0001$). Within the agnostic framework of §5.1, this evidence favors the measurement attenuation interpretation but does not exclude construct divergence.

L Formal Power Analysis

Table 13 reports the sample size required for Likert ATE to achieve 80% power at observed effect sizes across all six judge×axis cells. Required sample sizes range from 336 (Qwen3-32B verbosity, $d = 0.153$) to 27,159 (Qwen3-32B formality, $d = -0.017$), representing 4–206× the current n. For five of six cells, the required n exceeds 1,200, far beyond typical evaluation budgets, con-

Table 12: Bootstrap attenuation analysis: formality × Qwen3-32B. Bootstrap median Likert d remains flat near zero across sample sizes while BT is stable at 67.5%. Point estimate at $n = 355$: $d = -0.025$ (Table 1).

n	Likert d	95% CI	Contains 0?
50	−0.017	[−0.308, 0.262]	Yes
100	−0.015	[−0.219, 0.176]	Yes
150	−0.019	[−0.179, 0.145]	Yes
200	−0.016	[−0.157, 0.119]	Yes
250	−0.018	[−0.144, 0.107]	Yes
300	−0.018	[−0.131, 0.095]	Yes
355	−0.017	[−0.123, 0.087]	Yes

verging with the bootstrap attenuation curve (Appendix K) to confirm the Blindspot as a structural measurement limitation.

Table 13: Required Likert sample size for 80% power at observed d. BT detects all six effects at $p < 0.001$ with $n \leq 100$.

Judge	Axis	Observed d	Required n	Ratio
GPT-4o	Form.	0.073	1,473	16.7×
GPT-4o	Verb.	0.080	1,227	14.3×
Qwen3	Form.	−0.017	27,159	76.5×
Qwen3	Verb.	0.153	336	3.9×
Llama-70B	Form.	0.021	18,157	206.3×
Llama-70B	Verb.	0.034	6,737	78.3×

M Formality Sensitivity Hierarchy

Table 14 applies seven measurement methods to the same formality style-transformed data (GPT-4o, LLMBar, $n = 88$). Only pairwise methods reach significance; all Likert-based methods fail regardless of scaling assumption.

Table 14: Sensitivity hierarchy: seven measurement methods applied to the same formality data. Only pairwise methods (top two rows) reach significance.

Method	Statistic	p	Data
BT model	63.0% [55.7, 70.3]	0.0004	All 173 comp.
Sign test (filtered)	73.9% (34/46)	0.006	Consistent
Likert ATE (t-test)	$d = 0.102$	0.332	NLI≥0.90
Binarized Likert	60.6%	0.296	Non-tied
Ordinal reg. (prop. odds)	$\beta = -0.079$	0.769	NLI≥0.90
GEE (exchangeable)	$\beta = -0.084$	0.587	NLI≥0.90
Wilcoxon signed-rank	$W = 274$	0.905	NLI≥0.90

N Verbosity Sensitivity Hierarchy

Table 15 replicates the seven-method sensitivity analysis (Table 14) on the verbosity axis ($n = 86$ NLI-filtered pairs, Qwen3-32B judge, LLMBar). The attenuation pattern is more extreme than formality: all six Likert-based methods remain non-significant (lowest $p = 0.213$) despite a stronger pairwise signal (BT 77.3% vs. 66.5% for formality), and Likert effect sizes ($d = 0.036$–0.056) are near zero.

Table 15: Sensitivity hierarchy for verbosity ($n = 86$, Qwen3-32B, LLMBar). All Likert methods fail; only BT reaches significance.

Method	Statistic	p	Sig?
BT model	77.3% [70.3, 84.3]	<0.001	Yes
Likert ATE	$d = 0.056$	0.691	No
Binarized χ^2	OR = 2.16	0.213	No
Ordinal reg.	$\beta = 0.159$	0.575	No
Paired t-test	$d = 0.056$	0.607	No
Mann-Whitney U	$r = -0.046$	0.578	No
Cohen's d	$d = 0.036$	—	No

Effect sizes use style-aligned scoring (verbose − concise), isolating style preference direction. Table 1 reports RATE-compatible ATE (counterfactual − original), which conflates style and originality effects.

N.1 Scaled Replication ($n = 330$)

Table 16 reports verbosity results at $n = 330$ (from 355 generated pairs, 330 passing NLI gate at $\tau = 0.90$), replicating across four judges (GPT-4o was not re-scored at this scale). The Blindspot partially resolves for judges with strong style effects: Qwen3-32B ($d = .149$, $p < .001$) and Llama-70B ($d = .160$, $p < .001$) now reach Likert significance, consistent with the power curve (Figure 4) predicting ∼40% power at $n = 330$ for $d \approx 0.15$. However, two judges maintain the Blindspot: Gemini-2.5-Flash ($d = .087$, borderline) and Claude Sonnet 4 ($d = .032$, NS). The magnitude discrepancy persists even where Likert reaches significance: BT detects 73–76% win rates while Likert yields $d \approx 0.15$, confirming the attenuation interpretation.

GEE conditional decomposition at $n = 330$ confirms substantial shrinkage from $n = 88$ estimates, consistent with more extreme initial estimates producing larger corrections (Nemes et al., 2009): Qwen3 13.10 → 3.83 (−71%), Llama 8.21 → 4.01 (−51%), Gemini 8.98 → 2.52 (−72%). Despite shrinkage, verbosity remains style-dominated for these three judges (Style OR 2.5–4.0, all $p < .001$; Orig OR 0.52–0.69). Claude shows a non-significant style effect at $n = 330$ (Style OR = 1.22, $p = .134$), with its elevated BT reflecting originality rather than style preference (Orig OR = 1.69). As predicted by the power analysis (Figure 4), Likert begins detecting the strongest effects at $n \approx 330$ (Qwen3 $d = .15$, $p < .001$; Llama $d = .16$, $p < .001$), while two judges maintain the Blindspot (Claude $d = .03$, NS; Gemini borderline $d = .09$, $p = .02$ by Wilcoxon but $p = .11$ by t-test). The efficiency gap (Likert requiring 3–4× the sample size pairwise needs), rather than absolute failure, is the practical concern.

Table 16: Verbosity scaled replication ($n = 330$, four judges; GPT-4o not re-scored). Blindspot partially resolves for strong effects (Qwen3, Llama-70B) but persists for weaker-effect judges (Gemini, Claude).

Judge	BT P(V)	[95% CI]	Likert d (p)	Blindspot?
Qwen3-32B	.733***	[.696, .770]	.149 (.001)	No
Llama-70B	.759***	[.722, .793]	.160 (<.001)	No
Gemini-2.5	.678***	[.638, .717]	.087 (.021)	†
Claude S4	.608***	[.567, .649]	.032 (.317)	†

*** $p < 0.001$. † marks Blindspot cases (BT significant, Likert d non-significant at $\alpha = 0.05$ by Wilcoxon). Likert p values from Wilcoxon signed-rank test on non-zero differences.

O 100-Point Continuous Likert Scale

To test whether the Blindspot reflects integer-scale granularity rather than a paradigm difference between absolute and relative evaluation, we replicate the formality experiment using a 100-point continuous scale (1–100) with three judges (Table 17). All three judges remain non-significant (all $p > 0.15$). Tie rates decrease substantially with finer granularity (1–10 scale: 58–73%; 1–100 scale: 40–66%), yet ATE remains non-significant, confirming the gap reflects measurement mode rather than scale granularity (Preston and Colman, 2000). Judges differ markedly in effective granularity usage: GPT-4o clusters scores into ∼5 distinct bins (modes at 70, 75, 80), Qwen3-32B uses ∼20 distinct values spanning 45–95, and Llama-70B exploits 50+ unique scores with near-uniform coverage, yet none resolves the Blindspot. Score distributions are unimodal and right-skewed for all three judges, with medians in the 72–80 range; the score *difference* distributions (original minus rewrite) are symmetric around zero, confirming no net directional shift under Likert scoring.

P NLI Gate Ablation

Table 18 compares BT estimates across NLI-pass, NLI-fail, and all pairs. For formality, NLI-fail pairs reverse to 41.7% (preferring casual over formal), demonstrating that the gate removes content-confounded pairs rather than amplifying style signal. The small NLI-fail sample ($n \approx 12$) limits precision; this serves as a bounding analysis. For verbosity, NLI-fail pairs remain at 75.0%, consistent with the dominant style effect (OR = 13.10) overwhelming content noise.

Table 17: Formality Likert ATE: 1–10 vs. 1–100 scale. Finer granularity reduces tie rates but does not resolve the Blindspot (all 1–100 $p > 0.15$).

Judge	1–10 d (p)	1–100 d (p)	Tie 1–10	Tie 1–100
GPT-4o	—	−.038 (.725)	—	65.9%
Qwen3-32B	.012 (.908)	.153 (.154)	72.7%	51.1%
Llama-70B[a]	.000 (1.00)	.085 (.440)	72.7%	39.8%

[a] 5 pairs returned score = 0 (out of range); effective $n = 83$.

Table 18: NLI gate ablation: BT win probability by NLI subset.

Axis	NLI-pass	NLI-fail	All
Formality	66.5%	41.7%	63.0%
Verbosity	77.3%	75.0%	76.0%

Q Rewriting Artifact Control

Same-style double-rewrite (DR) controls quantify the rewriting artifact using BT models (consistent with all other pairwise analyses). Table 19 reports BT $P(\text{original})$, the probability the judge prefers the original over its same-style DR. Both judges systematically prefer LLM-polished text across axes; for verbosity, BT $P(\text{original}) < 0.17$ indicates that rewriting quality dominates when content is held constant, validating the GEE decomposition that separates style from artifact effects. Same-style double-rewrite controls for style-neutral polishing artifacts but cannot exclude style-specific polishing interactions (e.g., formal rewriting may polish differently than casual rewriting).

Table 19: Rewriting artifact: BT $P(\text{orig})$ vs. same-style double-rewrite (bootstrap 95% CI, 10,000 resamples). Values <0.5: judges prefer LLM-polished text.

Axis	Judge	BT $P(\text{orig})$ [CI]	p
Form.	GPT-4o	.418 [.342, .500]	.037
	Qwen3	.377 [.303, .451]	.0006
Verb.	GPT-4o	.167 [.084, .262]	<.0001
	Qwen3	.082 [.035, .140]	<.0001

R Implementation Details

Models and APIs. Style-controlled rewriting uses Qwen3-32B (temperature 0.7, top-p 0.9). Target judges: GPT-4o (`gpt-4o-2024-08-06`, accessed April–May 2026), Qwen3-32B (locally hosted), Llama-3.3-70B (locally hosted via vLLM). Claude Sonnet 4 (`claude-sonnet-4-20250514`, accessed May 2026) provides discriminative validity evidence with benchmark-dependent sensitivity. All judge calls use temperature 0 and `max_tokens` = 2048.

NLI gate. Content preservation is verified using DeBERTa-v3-large (He et al., 2023) fine-tuned on MNLI, with entailment threshold $\tau = 0.90$. Bidirectional entailment is applied for formality and register; direction-aware entailment for verbosity (Section 3.2).

Statistical methods. Bradley-Terry parameters are estimated via spectral initialization followed by minorization-maximization (Hunter, 2004) using the `choix` Python library (default L2 regularization $\alpha = 10^{-6}$); varying α from 10^{-6} to 1.0 changes all BT probabilities by <0.002. All bootstrap CIs use 10,000 pair-level resamples with seed 42. GEE models use exchangeable working correlation with robust (sandwich) standard errors via `statsmodels` (Section 5.2; diagnostics in Appendix D).

Benchmarks. LLMBar (Zeng et al., 2024): 100 instruction-response pairs ($n = 88$ after NLI gate, $n = 355$ for scaled replication). AlpacaEval (Dubois et al., 2025): 100 sampled pairs. MT-Bench (Zheng et al., 2023): 80 multi-turn pairs.

S Holm-Bonferroni Sensitivity Analysis

Table 20 reports adjusted p-values under the full 8-test family (4 judges × 2 primary axes), including Gemini-2.5-Flash which is excluded from the primary 6-test family (§4.1). All results remain significant at $\alpha = 0.05$.

T Cross-Axis Contamination Analysis

Our per-axis design assumes that rewriting along one style axis does not substantially alter the text along other axes. We quantify cross-axis contamination by measuring off-target surface proxies before and after rewriting. For formality rewrites ($n = 355$), we measure word count and sentence count as verbosity proxies; for verbosity rewrites ($n = 86$), we measure type-token ratio (TTR) as a lexical diversity proxy for formality, and Flesch-Kincaid (FK) grade level.

Table 20: Holm-Bonferroni sensitivity analysis: 8-test family (4 judges × 2 primary axes). All results remain significant under the expanded family ($\alpha = 0.05$).

Rank	Judge	Axis	BT P	Raw p	Threshold	Adj. p
1	Gemini Flash	Verb.	.708	<.0001	.00625	<.001
2	GPT-4o	Verb.	.760	.0001	.00714	.0007
3	Qwen3-32B	Form.	.665	.0001	.00833	.0007
4	Qwen3-32B	Verb.	.773	.0001	.0100	.0007
5	Llama-70B	Form.	.695	.0001	.0125	.0007
6	Llama-70B	Verb.	.735	.0001	.0167	.0007
7	GPT-4o	Form.	.630	.0004	.0250	.0008
8	Gemini Flash	Form.	.574	.0306	.0500	.0306

Table 21: Cross-axis contamination: effect of rewriting along one style axis on surface proxies of the other axis. Cohen's d: $|d| < 0.20$ negligible, 0.20–0.50 small contamination (Cohen, 1988).

Direction	Proxy	Δ	d	Interp.
Form. → Verb.	Word count	−3.4%	−0.23	Small
Form. → Verb.	Sent. count	−0.2%	−0.03	Negligible
Verb. → Form.	TTR	−0.006	−0.04	Negligible
Verb. → Form.	FK grade	—	0.50	Confounded*

* FK = f(sentence length, word length); sentence length change is inherent to verbosity manipulation, not formality contamination.

Formality rewrites leave sentence structure virtually unchanged ($d = -0.03$); the word count shift ($d = -0.23$, exceeding the negligible threshold but within the small range) reflects word-level adjustments (contractions, elaboration) that are directionally symmetric (formal→casual -5.0%, casual→formal $+5.6\%$) and do not constitute verbosity contamination. Verbosity rewrites leave lexical diversity unchanged (TTR $d = -0.04$); the elevated FK grade ($d = 0.50$) is an artifact of FK's dependence on sentence length, which changes by design in verbosity manipulation. These results confirm that cross-axis contamination is small to negligible for the proxies measured; the GEE originality term (§5.2) provides additional protection by absorbing any residual quality confound. A robustness check adding word-count change as a GEE covariate leaves all Style ORs virtually unchanged ($\Delta < 12\%$, all covariate $p > .28$), confirming that the small cross-axis word-count shift does not confound formality estimates.

U FK Length-Control Analysis

The elevated FK grade shift for verbosity rewrites ($d = 0.50$; Table 21) could reflect genuine formality contamination or simply FK's structural dependence on sentence length. We disentangle these via length-controlled regression and mediation analysis.

Table 22: Descriptive statistics for FK grade and sentence length changes by rewriting axis. Computed on all pairs; regression analysis below uses NLI-verified pairs only ($n_{\text{form}} = 88$, $n_{\text{verb}} = 86$).

Axis	Metric	Δ	d
Verbosity	FK grade	$+2.26$	0.50
Verbosity	Sent. length (words)	$+3.40$	0.35
Formality	FK grade	-0.30	-0.08
Formality	Sent. length (words)	-0.11	-0.04

Regressing FK change on style axis without controls yields $\beta = 2.69$ ($p < .001$, $R^2 = 0.100$); adding sentence length as a covariate reduces the coefficient to $\beta = 1.07$ ($p = .020$, $R^2 = 0.546$), a 60.3% reduction. Sobel mediation test confirms that sentence length mediates 60% of the total effect ($z = 2.15$, $p = .031$); the 40% residual direct effect reflects multi-syllable vocabulary shifts (e.g., "use" $\rightarrow$ "utilize") that are inherent to verbose style rather than formality contamination.

GEE with FK covariate. As a further robustness check, we add FK grade change as a covariate to the GEE formality model. FK grade is not independently significant for any judge ($p > .11$), but its inclusion reduces formality Style ORs by 20–51% due to collinearity between formality and readability: GPT-4o Style OR drops from 2.70 to 2.15 ($p = .132$, no longer significant); Qwen3-32B from 4.20 to 3.38 ($p = .014$); Llama-70B from 5.52 to 2.69 ($p = .037$). Two of three judges retain significant formality effects after readability control, confirming that formality style sensitivity is not fully reducible to readability differences.

V Human Annotation Agreement Statistics

Two independent annotators evaluated 100 NLI-scored pairs (50 formality, 50 verbosity), stratified into 70 NLI-pass and 30 NLI-fail pairs for content preservation. Annotator 1 (A1) judged 99/100 pairs content-preserving; Annotator 2 (A2) judged 97/100.

Agreement. On the *preserves_meaning* label, 96/100 pairs are concordant, yielding Gwet's $\mathrm{AC}_1 = 0.958$ (Gwet, 2008). Cohen's $\kappa = -0.015$ is misleading here: with >96% positive prevalence, chance agreement is near-ceiling and κ is compressed toward zero regardless of actual agreement, the well-documented prevalence paradox (Feinstein and Cicchetti, 1990). AC_1 is the appropriate statistic under such extreme marginal distributions.

Gate pass precision. Both annotators independently confirm 69/70 gate-passed pairs as content-preserving (98.6%; Wilson CI $[92.3\%, 99.8\%]$). Only two gate-passed pairs were flagged by either annotator (S073 and S063, both formality); excluding these changes BT by <0.3 pp and Style OR by <4% relative, with all significance levels unchanged.

Gate-failed analysis. Among 30 gate-rejected pairs, A1 judges 30/30 and A2 judges 28/30 as content-preserving, confirming that the gate's dominant error mode is false rejection rather than false acceptance. This conservative calibration ensures high pass precision at the cost of reduced recall (59.5%).

Disagreement analysis. Four pairs show annotator disagreement: S073, S098, S063, S136 (3 formality, 1 verbosity; 2 NLI-pass, 2 NLI-fail). All disagreements involve borderline meaning-equivalence judgments rather than clear content contamination.

Comparison with 50-pair pilot. The expanded 100-pair annotation tightens the precision estimate from 84–100% (individual annotators on 25 gate-passed pairs) to 98.6% (both annotators concordant on 70 gate-passed pairs), and raises AC_1 from 0.65 to 0.96.

Style detection agreement. Agreement on *style_change_detected* is lower (58%, $\kappa = 0.138$, $\mathrm{AC}_1 = 0.206$), reflecting genuine difficulty in judging whether a rewrite altered style. This does not affect the primary validation target, which concerns content preservation rather than style detection.

W Cross-Benchmark Replication

Table 23 reports cross-benchmark replication results on AlpacaEval and MT-Bench. Selection criterion: we targeted the six judge×axis cells with the largest LLMBar BT effects (BT $\geq .63$, all $p < .001$), prioritizing cells where the Blindspot was strongest. The remaining 18/24 cells (3 benchmarks $\times$ 4 target judges $\times$ 2 axes, minus 6 tested) were not evaluated; generalization to weaker-effect configurations remains open.

Table 23: Cross-benchmark replication on AlpacaEval (AEval) and MT-Bench (MTB). ‡: both BT and Likert significant; †: Blindspot (BT significant, Likert not).

Bench	Judge	Axis	BT P	Likert d (p)	Cat.
AEval	GPT-4o	Form.	.727***	.005 (.909)	†
	Qwen3	Form.	.657***	.081 (.198)	†
	Qwen3	Verb.	.841***	.375 (.0003)	‡
	Llama-70B	Form.	.750***	.212 (.012)	‡
MTB	GPT-4o	Form.	.692***	.111 (.016)	‡
	Qwen3	Form.	.712***	.149 (.007)	‡

*** $p < 0.001$. ‡: both sig.; †: Blindspot.

X Claude Cross-Benchmark Analysis

Claude Sonnet 4 exhibits near-null style sensitivity on LLMBar (§5.4; Appendix H). Cross-benchmark GEE decomposition reveals that Claude's elevated aggregate BT on AEval/MT-Bench reflects quality detection rather than style preference. Table 24 reports results across all five tested configurations.

Table 24: Claude Sonnet 4 cross-benchmark results with GEE decomposition. Formality Style OR is non-significant across all three benchmarks; elevated BT on AEval/MT-Bench is driven by Orig OR (quality detection). AEval verbosity is the sole genuine style effect.

Bench	Axis	BT P	Likert d (p)	Style OR (p)	Orig OR (p)
LLMBar	Form.	.562	−.021 (.721)	1.47 (.276)	6.66 (<.001)
LLMBar	Verb.	.541	−.039 (.743)	1.52 (.286)	0.80 (.563)
AEval	Form.	.698***	.081 (.090)	1.99 (.082)	9.33 (<.001)
AEval	Verb.	.683***	.252 (.002)	**4.76** (<.001)	0.87 (.732)
MTB	Form.	.733***	.233 (<.001)	1.64 (.364)	11.85 (<.001)

*** $p < 0.001$. All Style ORs except AEval verbosity are non-significant. Negative d: styled variant scored lower.

Formality. Claude's formality Style OR is non-significant across all three benchmarks (1.47/1.99/1.64, all $p > .08$), indicating consistently low formality sensitivity regardless of benchmark. The elevated BT on AEval (69.8%) and MT-Bench (73.3%) is driven entirely by Orig OR (9.33/11.85, both $p < .001$): Claude detects quality differences between original and counterfactual texts without exhibiting formality preference. AEval formality produces a Blindspot (BT significant, Likert n.s.), but GEE decomposition reveals this as quality detection rather than style sensitivity, illustrating why aggregate BT alone is insufficient for bias attribution.

Verbosity. AEval verbosity is the sole configuration where Claude exhibits genuine style sensitivity (Style OR $= 4.76$, $p < .001$; Orig OR $= 0.87$, n.s.), a style-dominated pattern. LLMBar verbosity remains null (Style OR $= 1.52$, n.s.), confirming benchmark-dependent sensitivity.

Y Rewriting Prompts

Below are the axis-specific rewriting prompts used in Section 3.1. All prompts are issued to Qwen3-32B-Instruct at temperature 0.7.

Formality. `Rewrite the following text to be more [formal/casual] while preserving the exact same content and meaning. Do not add, remove, or change any factual information. Only adjust the level of formality in the language.`

Verbosity. `Rewrite the following text to be more [verbose/concise] while preserving the exact same content and meaning. Do not add, remove, or change any factual information. Only adjust the level of detail and elaboration.`

Register. `Rewrite the following text to use a more [academic/conversational] register while preserving the exact same content and meaning. Do not add, remove, or change any factual information. Only adjust the register and tone.`

Same-style double-rewrite. `Rewrite the following text in the same style (keep it [formal/casual/verbose/concise]) while preserving the exact same content and meaning. Paraphrase the text without changing its style.`

Z Non-Tie Concordance Analysis

To adjudicate between measurement attenuation and construct divergence as explanations for the Likert Blindspot (§5.1), we examine whether Likert and pairwise evaluations agree *when Likert does differentiate*. For each pair where Likert assigned different scores to the two texts (non-tie), we check whether the higher-scoring text matches the pairwise-preferred text; if the two methods measure the same underlying construct, non-tie concordance should substantially exceed chance (50%).

We additionally compute point-biserial correlations (r_{pb}) between Likert score differences and pairwise outcomes across all pairs with valid pairwise preferences.

Table 25: Non-tie concordance between Likert and pairwise preferences. For each judge×axis condition, we identify Likert non-tie pairs and check agreement with pairwise preference direction. Wilson 95% CIs; p from two-sided binomial test against 50%. r_{pb}: point-biserial correlation between Likert score difference and pairwise outcome (all pairs with valid pairwise preference).

Bench	Config	n_{nt}	Conc.	Rate	95% CI	p	r_{pb} (p)
LLMBar	Llama verb.	25	22	88.0%	[70.0, 95.8]	<.001	.412 (.001)
	GPT-4o verb.	30	25	83.3%	[66.4, 92.7]	<.001	.373 (.003)
	Llama form.	18	15	83.3%	[60.8, 94.2]	.008	.447 (.001)
	Qwen3 verb.	27	20	74.1%	[55.3, 86.8]	.019	.296 (.019)
	Gemini verb.	24	17	70.8%	[50.8, 85.1]	.064	.346 (.012)
	Qwen3 form.	15	10	66.7%	[41.7, 84.8]	.302	.274 (.065)
	Gemini form.	15	10	66.7%	[41.7, 84.8]	.302	.373 (.012)
AEval	Qwen3 form.	26	20	76.9%	[57.9, 89.0]	.009	.203 (.146)
	GPT-4o form.	21	15	71.4%	[50.0, 86.2]	.078	.340 (.019)
Pooled		**201**	**154**	**76.6%**	**[70.3, 81.9]**	**<.001**	—

Pooled concordance is 76.6% (154/201 non-tie pairs; binomial $p < .001$; Table 25), substantially exceeding the 50% chance baseline. Point-biserial correlations are positive in all nine conditions ($r_{pb} = 0.20$–0.45), with seven of nine reaching significance ($p < .05$). Concordance is highest for verbosity (74–88%) and for Llama-70B across both axes (83–88%), consistent with Llama's lower Likert tie rate (Table 17).

These results favor the measurement attenuation interpretation: when Likert scoring does differentiate between texts, its preference direction aligns with pairwise judgment in more than three-quarters of cases. The primary driver of the Blindspot is thus the high Likert tie rate (60–73% across conditions) rather than a fundamental disagreement between the two evaluation paradigms about what constitutes quality. The $\sim$23% discordance rate leaves room for partial construct divergence (pairwise comparison may access quality dimensions that Likert cannot express), but the dominant pattern is signal attenuation through discretization and ceiling effects.

Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform the related-work section.